

LoRA as Declarative Adaptation for Multimodal Clinical Question Answering in a Virtual Hospital Environment

Sandrine de Ribaupierre^{*†}, Roy Eagleson^{†‡}, Anthony Barros[‡], Kelsey Kloosterman[‡]

^{*}Western University CNS [†]Centre for Brain and Mind, Western University

[‡]Faculty of Engineering, Dept. of AI and Software Engineering, Western University

Abstract—Pediatric patients in hospital environments require age-appropriate, institutionally grounded information that general-purpose AI systems are not equipped to provide. We present a LoRA fine-tuning framework for clinical question-answering within a virtual hospital environment, targeting two pediatric age groups (5–11 and 12–18). Using Low-Rank Adaptation (LoRA) [8] on 4-bit quantized Llama 3.2 Instruct models (1B and 3B), we train Standard ($r=8$, 16 layers) and Fast ($r=4$, 8 layers) adapter configurations, producing eight training runs across model sizes and age groups. Fine-tuned adapters substantially outperform zero-shot base models on age-appropriate readability [1], [2], [3], [4], [5], [6], [7] (Flesch-Kincaid Grade ≤ 7.0 pass rates: 72.84% vs. 12.14%) and inter-role style separation (0.89–0.94 vs. 0.66–0.70). We observe a configuration-ordering crossover between 1B and 3B models: the Standard adapter outperforms on 3B (84% vs. 72%), while Fast outperforms on 1B (82% vs. 76%), consistent across all five readability metrics. Only Fast-1B meets the 1.0 s VR latency target. These results demonstrate that compact LoRA adaptation on consumer hardware is feasible for pediatric clinical NLP, but adapter configurations do not transfer reliably across model scales [10].

Index Terms—Clinical Q&A systems, Pediatric NLP, LoRA, Virtual Reality

I. INTRODUCTION

We present results situated within a virtual hospital framework previously developed by our group to support interactive, clinically grounded communication among pediatric patients, caregivers, and clinical teams. In this work, we extend the platform with a constrained multimodal question-answering capability integrating large language models (LLMs) and vision-language models (VLMs) to enable safe, naturalistic dialogue aligned with institutional guidance and clinical workflows. Our primary objective is to determine whether Low-Rank Adaptation (LoRA) [8] can produce quantifiably age-appropriate pediatric question answering [1], [2], [3], [4], [5], [6], [7] with real-time local deployment on commodity graphical workstations, and how optimal adapter configurations depend on model scale.

II. MATERIALS AND METHODS

Using Low-Rank Adaptation (LoRA) [8] on 4-bit quantized Llama 3.2 Instruct models (1B and 3B) [9], we locally train two adapter configurations: Standard ($r=8$, 16 layers) and Fast ($r=4$, 8 layers), across both model sizes and age groups for 8 training runs total. Training data consists of 500 synthetic examples per adapter, generated by Claude Opus 4.6

using Victoria Hospital’s public patient materials as grounding artifacts, following a LIMA-inspired curation approach [11].

We conceptualize LoRA as a form of *declarative adaptation*: structured institutional knowledge is encoded as modular behavioral constraints rather than procedural rules. Hospital guidelines, care pathways, and patient information resources define a specification space of acceptable responses, while the adapted model realizes that specification during interaction. We further introduce a five-factor corpus quality framework—correctness, consistency, completeness, conclusiveness, and relevance—used both to construct controlled synthetic training corpora and to evaluate downstream responses.

III. RESULTS

Fine-tuned adapters substantially outperform zero-shot base models on age-appropriate readability [1], [2], [3], [4], [5], [6], [7] (Flesch-Kincaid Grade 7.0 pass rates: 72.84% vs. 12.14%) and inter-role style separation (0.89–0.94 vs. 0.66–0.70). A notable empirical finding is a **configuration-ordering crossover** across model scales [10]. Under a Standard configuration, 3B models outperform 1B on readability (84% vs. 72% pass rate); under a Fast configuration this ordering reverses, with Fast-1B outperforming Fast-3B (82% vs. 76%). This crossover is directionally consistent across all five readability measures examined. Latency results indicate that only Fast-1B satisfies the sub-second response target required for immersive VR interaction.

IV. DISCUSSION AND CONCLUSION

We evaluated the resulting system in a medical visual question-answering setting, where users pose natural language queries regarding procedures, imaging workflows, expected duration, comfort measures, and care safeguards. Performance was assessed through automated readability metrics [1], [2], [3], [4], [5], [6], [7], inter-role style separation, latency measurements, and expert inspection. Our findings suggest that compact LoRA adaptation [8] provides a practical pathway for privacy-preserving pediatric clinical NLP on consumer hardware, but adapter configurations do not transfer reliably across model scales [10]. Combining synthetic corpus design [11], multimodal adaptation, and local deployment offers a scalable and governable route toward clinical communication systems in virtual healthcare environments.

Disclosures: The authors declare no conflict of interest.

REFERENCES

- [1] R. Flesch, "A new readability yardstick," *J. Applied Psychology*, vol. 32, no. 3, pp. 221–233, 1948.
- [2] R. Gunning, *The Technique of Clear Writing*. New York: McGraw-Hill, 1952, pp. 36–37.
- [3] R. J. Senter and E. A. Smith, "Automated readability index," AMRL-TR-66-20, Wright-Patterson AFB, 1967.
- [4] G. H. McLaughlin, "SMOG grading—a new readability formula," *J. Reading*, vol. 22, pp. 639–646, 1969.
- [5] J. P. Kincaid and W. C. McDaniel, "An inexpensive automated way of calculating Flesch Reading Ease scores," Patent Disclosure 031 350, U.S. Patent Office, 1974.
- [6] J. P. Kincaid, L. Fishburne, R. Rogers, and B. Chissom, "Derivation of new readability formulas for Navy enlisted personnel," Research Branch Report 8-75, Naval Technical Training Command, 1975.
- [7] M. Coleman and T. L. Liao, "A computer readability formula designed for machine scoring," *J. Applied Psychology*, vol. 60, pp. 283–284, 1975.
- [8] E. J. Hu, Y. Shen, P. Wallis, et al., "LoRA: Low-Rank Adaptation of large language models," in *Proc. ICLR*, 2022.
- [9] T. Detrmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Adv. Neural Inf. Process. Syst.*, 2023.
- [10] D. Biderman, J. G. Ortiz, J. Portes, et al., "LoRA learns less and forgets less," *Trans. Mach. Learn. Res.*, 2024.
- [11] C. Zhou, P. Liu, P. Xu, et al., "LIMA: Less is more for alignment," in *Adv. Neural Inf. Process. Syst.*, 2023.